

Trans, bar: DS + Algorithms

Proofs &
Sets



Recap

Worksheet - how did it go?

Next 3 readings posted

HW1 - coming!

Last time: Proofs

Two way implications

Example: For any $n \in \mathbb{Z}$,
 n is odd $\iff n^2$ is odd.

- Here, $\iff$ reads as "if and only if"
- Sometimes written $p \iff q$
so $(p \rightarrow q) \wedge (q \rightarrow p)$

How to prove:

Let's try:

For any $n \in \mathbb{Z}$,
 n is odd $\Leftrightarrow n^2$ is odd.

Proof:

$\Rightarrow$:

$\Leftarrow$:

Proof by contradiction: (or indirect proof)

Recall that a contradiction is a logical statement which is always false

Ex: $2=1$
 $x = x+1$

} F

Consider $p \rightarrow q$:

P	q	$p \rightarrow q$
T	T	
T	F	
F	T	
F	F	

Bad row: want to exclude.

so: $p \wedge \neg q$ is thing to avoid!

Proof by contradiction:

Assume $P \wedge \neg Q$, then show some logical contradiction exists.

Example: If n^2 is even, then n is even.

Another:

Prove that $\sqrt{2}$ is irrational.

Recall: x rational means

$$\exists p, q \in \mathbb{Z} \text{ s.t. } x = \frac{p}{q}$$

Proof: assume $\sqrt{2}$ is rational,
+ then make a contradiction

Sets

Unordered collections.

Vs ordered sets (or tuples)

Ex: $\{1, 3, 5, 7\} =$

$\emptyset$

$\{1, 2, 3, \dots, 100\}$

$\{n^2 : n \in \mathbb{N}\}$

$\{\emptyset, \{1, 2\}, \{1\}, \{2\}\}$

Cardinality & repetition:

For any set, we don't count with multiplicity.

$$\text{So: } \{x : 2|x \text{ or } 3|x\}$$

The size of a set $S =$
of distinct elements.

$$\text{So } |\{1, 2, 3, 2, 5, 1\}| =$$

Subsets

$$X \subseteq Y \stackrel{\text{def}}{\iff} \forall x, x \in X \rightarrow x \in Y.$$

Examples:

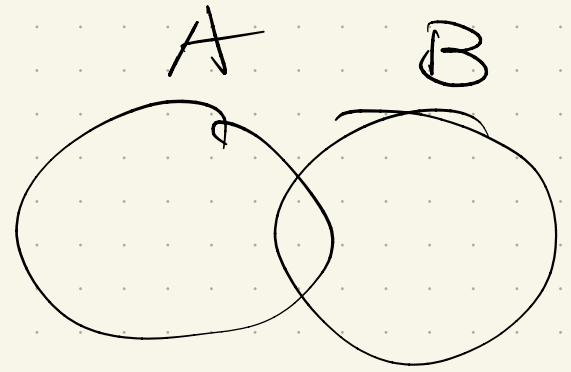
$$\bullet \{1, 3, 5\} \subseteq \{1, 2, 3, \dots, 100\}$$

$$\bullet \emptyset \subseteq \{1, 2, 3\}$$

Why?

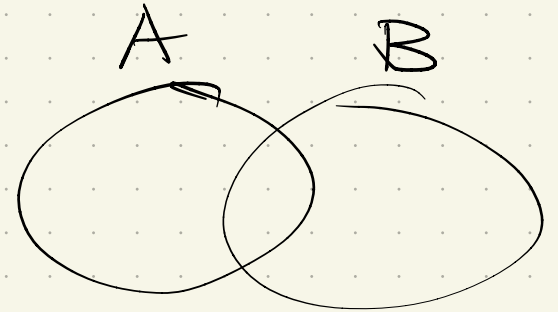
Union:

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$



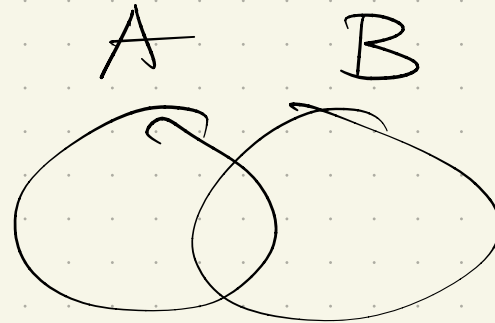
Intersection

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



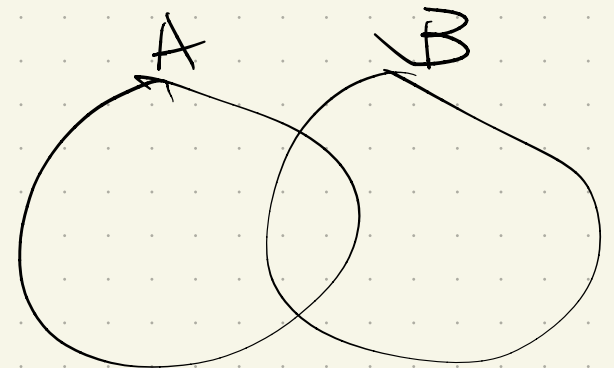
Set difference

$$A - B \text{ (or } A \setminus B) = \{x \mid x \in A \wedge x \notin B\}$$



Symmetric difference

$$A \oplus B = (A - B) \cup (B - A)$$



DeMorgan (* operations)

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \quad * \quad \overline{A \cap B} = \bar{A} \cup \bar{B}$$

How to prove?

First, complement $\bar{A} = \{x \mid x \notin A\}$

① Take $x \in \overline{A \cup B}$, show in $\bar{A} \cap \bar{B}$

② Take $x \in \overline{A} \cap \overline{B}$

Those Laws!

Name	Identities	
Idempotent laws	$A \cup A = A$	$A \cap A = A$
Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Commutative laws	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Distributive laws	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity laws	$A \cup \emptyset = A$	$A \cap U = A$
Domination laws	$A \cap \emptyset = \emptyset$	$A \cup U = U$
Double complement law	$\overline{\overline{A}} = A$	
Complement laws	$A \cap \overline{A} = \emptyset$ $\overline{\overline{U}} = U$	$A \cup \overline{A} = U$ $\overline{\emptyset} = U$
De Morgan's laws	$\overline{A \cup B} = \overline{A} \cap \overline{B}$	$\overline{A \cap B} = \overline{A} \cup \overline{B}$
Absorption laws	$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$

Cartesian Products

Sometimes, order does matter!

Tuples or strings!

Formalize: Given 2 sets A and B , the product of A and B , written $A \times B$, is the set of ordered pairs where the first element is from A and the second from B .

$$A \times B = \{ (a, b) : \quad \}$$

Examples:

Bit strings:

Strings:

$\mathbb{R}^2$ or $\mathbb{R}^3$:

Cauton

$$(A \times B) \times C \neq A \times B \times C$$

Element of $(A \times B) \times C$:

element of $A \times B \times C$:

Another

What is $\phi \times \{a, b\}$?

